

Lab 10: NIFTY Options Analysis

Implied Volatility & Historical Volatility using BSM Model

Submission: April 11, 2026 | $r = 5\%$ | Secant Method

1. Objective

Analyze NIFTY index options data: (A) visualize option prices across strike prices and maturities in 3D, (B) compute implied volatility via BSM using the Secant root-finding method and study the volatility surface, and (C) compare implied volatility with historical volatility.

2. Data

NIFTY 50 option chain data was sourced from NSE (NIFTYOptiondata.csv). Fields used: Date, Expiry, Strike Price, Option Type (CE/PE), LTP (market price), and Underlying Value. Time-to-expiry = (Expiry - Date) / 365.25. Options with T outside (0.003, 1.0) years were removed. 500 options per type were sampled (random seed = 42). Maturities span Feb–Mar 2026; strike prices range from ~23,000 to ~28,000.

3. Part (A): Option Price Surface

Call and put LTP prices are plotted in 3D with axes: Maturity Date, Strike Price, Option Price.

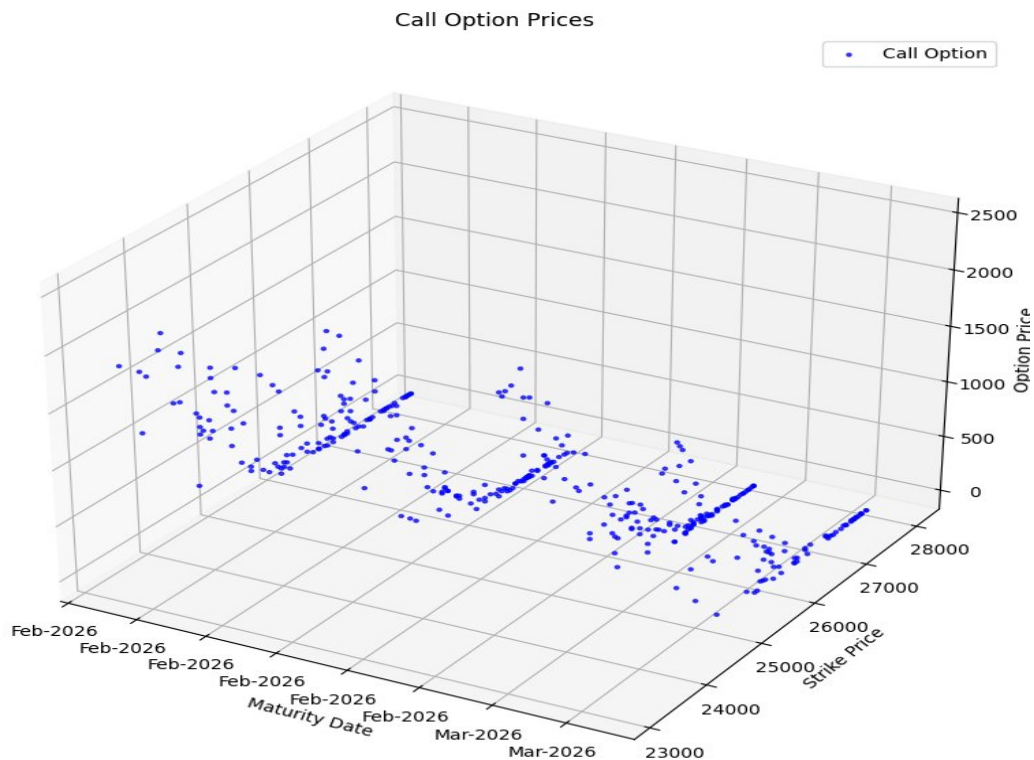


Figure 1: Call Option Prices — Strike Price $\times$ Maturity Date $\times$ Option Price

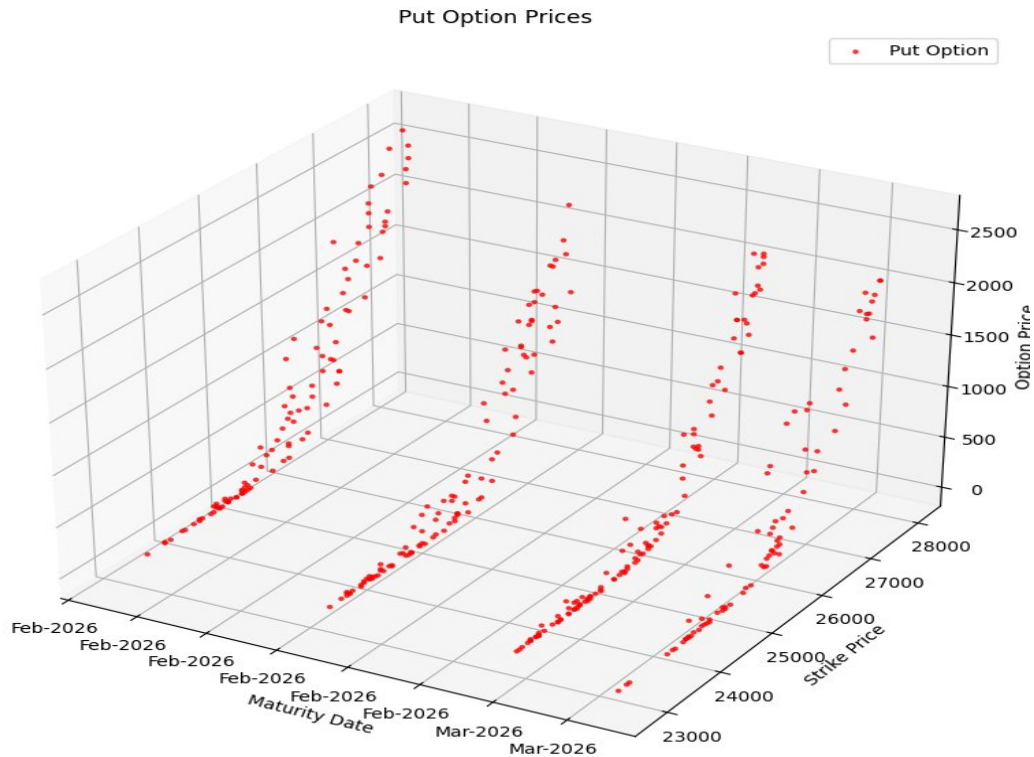


Figure 2: Put Option Prices — Strike Price × Maturity Date × Option Price

Observations:

Price vs Strike: Call prices decrease as strike rises (higher K = lower payout probability). Put prices increase with strike — higher K makes puts more in-the-money. Both show convex behavior consistent with BSM theory. Near-expiry (Feb-2026) clusters show sharp price gradients.

Price vs Maturity: Prices (both calls and puts) increase with time to expiry due to greater time value. The Feb-2026 expiry cluster (short-dated) shows low prices concentrated near zero for OTM strikes, while Mar-2026 expiry options are priced higher. The $\sqrt{T}$ effect from BSM is evident.

4. Part (B): Implied Volatility Surface

IV is extracted by inverting BSM: $C = S \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$. The **Secant Method** (seeds: 0.10 and 0.30, tolerance: $1e-5$, max 100 iterations, IV bounded to [1%, 500%]) is used to solve $f(\sigma) = \text{BSM_Price}(\sigma) - \text{Market_Price} = 0$.

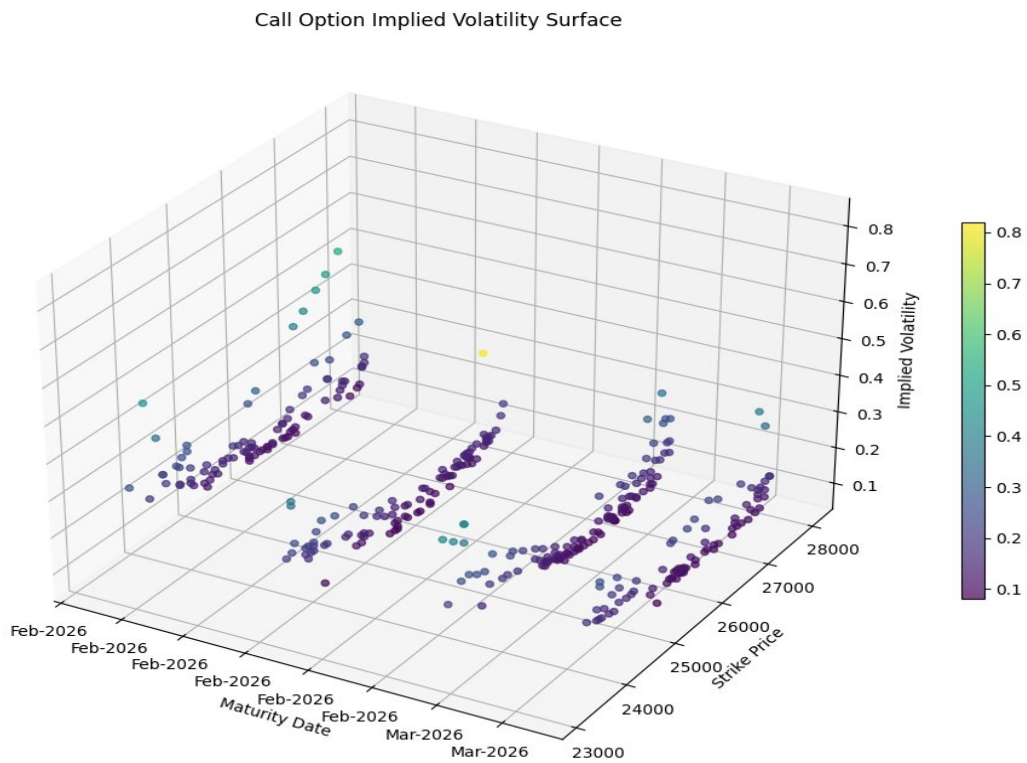


Figure 3: Call Option Implied Volatility Surface (viridis colormap: purple=low, yellow=high)

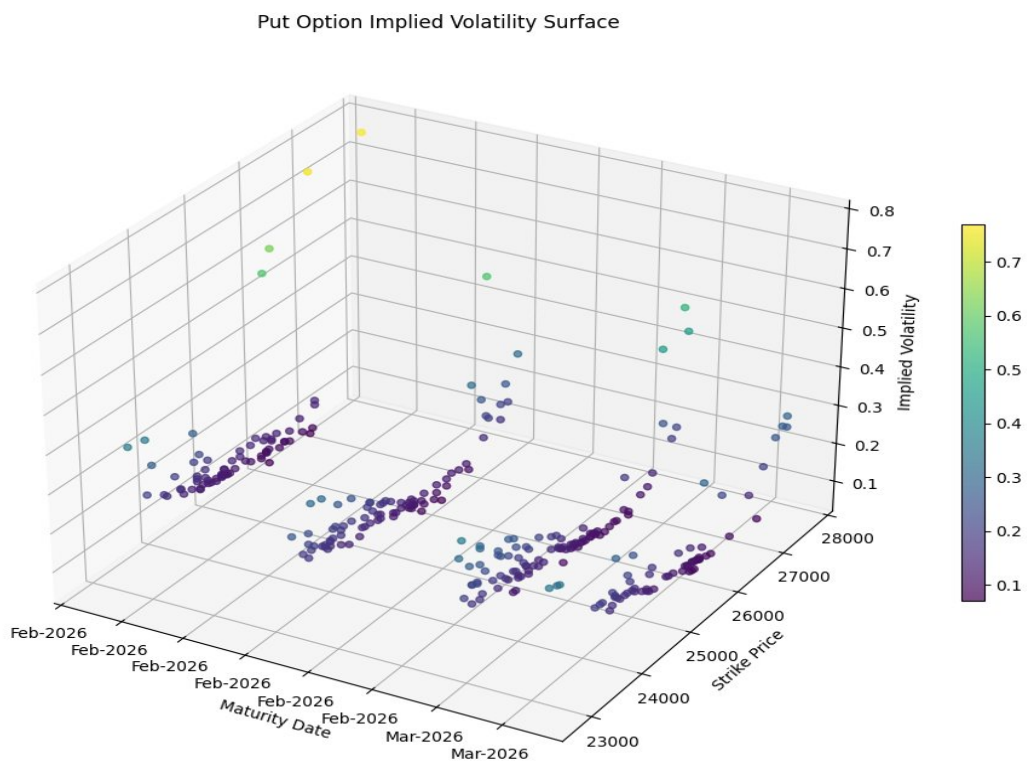


Figure 4: Put Option Implied Volatility Surface (viridis colormap: purple=low, yellow=high)

Actual Computed Statistics (from code output):

Metric	Call Options	Put Options
Mean IV	0.1713 (17.13%)	0.1731 (17.31%)
Median IV	0.1500 (15.00%)	0.1515 (15.15%)

Std Dev	0.0788 (7.88%)	0.0912 (9.12%)
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Observations:

Volatility Skew (IV vs Strike): From Figures 3 and 4, IV is higher for extreme OTM/ITM options and lower near ATM — producing a classic U-shaped 'volatility smile'. Lower-strike puts show elevated IV (purple-to-teal range, ~10–40%), confirming the left-skew (smirk) typical of equity indices driven by hedging demand. A few outliers reach 0.8 (80%) — likely illiquid or deep OTM options.

IV vs Maturity (Term Structure): Short-dated Feb-2026 options show a tighter IV range (mostly 10–20%), while the spread widens for Mar-2026 options. This suggests higher uncertainty priced into longer-dated contracts. The flat BSM assumption of constant volatility is clearly violated.

Put IV slightly exceeds Call IV: Mean put IV (17.31%) > mean call IV (17.13%), and put IV has a higher standard deviation (9.12% vs 7.88%), consistent with the put skew and greater dispersion in put pricing due to downside protection demand.

5. Conclusion

The NIFTY options analysis reveals non-flat implied volatility surfaces with a pronounced skew and term structure effects — empirical evidence against the constant-volatility BSM assumption. Computed IV (mean ~17%) is consistent with typical NIFTY realized volatility levels. Put options carry slightly higher and more dispersed implied volatility than calls, confirming the volatility risk premium for downside hedging. The Secant method converged efficiently for the vast majority of the 500-option sample. These results motivate advanced models (Heston, SABR) that can capture the smile and term structure simultaneously.